DE–4143 **11**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

ORGANIC CHEMISTRY – I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Use the Huckel rule to indicate wheather the followings are aromatic or antiaromatic explain.

   (i) Cyclopentadienyl anion

   (ii) Ferrocene

   (iii) Azulene

   (iv) Annulene (4 × 2 = 8)

   (b) Using an energy level profile diagram explain the terms kinetic control and thermodynamic control of the reactions with suitable examples. (6)

   (c) Write a note on,

   (i) Steric inhibition of resonance

   (ii) Steric enhancement of resonance

   (iii) Mesomeric effect. (3 × 2 = 6)

2. (a) What are nitrenes? Explain their formation and stability. (6)

   (b) State and explain Hammond postulate. (4)

(c) Explain alternant and non-alternant hydrocarbons with suitable examples. (4)

(d) Give S_N2 mechanism. Discuss the solvent and substrate effects. (6)

3. (a) Discuss briefly S_E1 and S_Ei mechanisms. (8)

(b) Explain the following

(i) Vilsmier Haak reaction

(ii) Phase transfer catalysis

(iii) Hard and soft acids

(iv) Benzyne mechanism. (4 × 3 = 12)

4. Write the mechanism of following reaction:

(a) Perkin reaction

(b) Reformatsky reaction

(c) Claise ester condensation

(d) Dickmann condensation

(e) Benzoin condensation. (5 × 4 = 20)

5. (a) Write a short notes on, Regio selectivity and chemo selectivity. (4)

(b) Discuss briefly E_2 and E_1 CB mechanisms. (8)

(c) Explain

(i) Hydroboration

(ii) Hydration of olefins

(iii) Michael addition

(iv) Bredt's rule. (4 × 2 = 8)

6. Draw the structure and give the synthesis of following compounds:

(a) Indole

(b) Morphine

(c) Atropine

(d) Flavone

(e) α-pinene. $(5 \times 4 = 20)$

7. (a) How is the structure of camphor established? (8)

(b) Write down the biosynthesis of alkaloids. (4)

(c) Write down the biological importance of DNA and RNA. (8)

8. (a) Establish the structure of starch and cellulose. (10)

(b) Write a note on,

(i) Solid phase peptide synthesis

(ii) End group analysis. $(2 \times 5 = 10)$

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

INORGANIC CHEMISTRY–I

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carriers 20 marks.

(5 × 20 = 100)

1. (a) Predict the structure of SCl_2 and $SnCl_2$. (8)

(b) Derive Born-Lande equation for the lattice energy of an ionic compound. (7)

(c) State and explain HSAB concept. Give a critical account on its applications. (5)

2. (a) How is F centre formed? (4)

(b) Draw the structure of zinc blende and wurtizite and show the coordination around each type of ion in these structures. (8)

(c) Give the conditions for the formation of frenkel and schottky defects. Explain the consequences of these two defects. (8)

3. (a) Compute Q for Li^7 (p, α) He^4 given that the atomic masses for Li^7, p, and α are respectively 7.01822, 1.00814 and 4.00387. (4)

(b) Discuss the principles and applications of neutron activation analysis. (8)

(c) Explain how fission and spallation reactions differ each other. Discuss the liquid drop model for fission reactions. (8)

4. (a) What is shift reagent? Give an example. (4)

(b) Discuss the ion exchange method of the separation of lanthanides. (8)

(c) Compare the spectral and magnetic properties of lanthanides and actinides. (8)

5. (a) What are garnet minerals? Give an example. (4)

(b) Name the isopoly acids formed by vanadium and molybdenum. (8)

(c) How is 12-phosphotungstic acid prepared? Discuss its structure. (8)

6. (a) What are molecular sieves? Give an example. (4)

(b) Explain photovoltaic effect and its applications. (8)

(c) Discuss the construction and working principle of (i) Cloud chamber and (ii) Cyclotron. (8)

7. (a) Would you expect both Al_2Cl_6 and $Al(CH_3)_6$ to be electron deficient compounds? Give reason. (4)

(b) Discuss the position of lanthanides and actinides in the periodic table. (8)

(c) Give an account on nuclear structure and nuclear forces. (8)

8. (a) How will you identify the p or n type semiconductors? (5)

(b) Calculate U_o for KB_r from the following data in Kcal/mole.

$\Delta H_{AM} = 21$; $\Delta H_{AX} = -80$; IE=99; DE=46 $\Delta H_f = -97$ (5)

(c) Compare and contrast VB and MO approaches of bonding in covalent compounds. (10)

DE–4145 **13**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012

PHYSICAL CHEMISTRY – I

(2008 onwards)

Time : 3 Hours Maximum : 100 Marks

Answer any FIVE questions.

Each questions carries 20 marks. ($5 \times 20 = 100$)

1. (a) Derive any two of the Maxwell's relations. (10)

   (b) What is meant by fugacity? How is the mean activity coefficient of an electrolyte determined? (10)

2. (a) Derive Debye Huckel onsager equation. (10)

   (b) Write briefly about electrode kinetics over voltage and how it could be determined. (10)

3. (a) Explain about wave particle dualism with a suitable example and highlight about the inadequacy of classical mechanics. (10)

   (b) Discuss the particle in three dimensional box problems, derive the solution. (10)

4. (a) Give the thermodynamic and statistical treatment for absolute Reaction Rate Theory. (12)

   (b) Write short notes on kinetics of complex reactions. (8)

5. (a) Describe the different photophysical and photochemical processes of the electronically excited states. (14)

(b) Write briefly about Flash photolysis. (6)

6. (a) Outline the significance of equilibrium constant and its dependence on temperature and pressure. (6)

(b) Write notes on the following :

(i) Transport number and ionic mobilities. (4)

(ii) Glass electrode. (4)

(iii) Nernst equation and its significance. (6)

7. (a) What is meant by degeneracy? (4)

(b) Describe about Lindemann, Hinshelwood, KRRM and slater theory in chemical kinetics. (16)

8. (a) Give the importance of steady state approximation. (6)

(b) Write briefly about hydrogen-halogen reaction and decomposition of carbonyl compounds. (14)

DE–4146 **14**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

INSTRUMENTAL METHODS OF ANALYSIS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Draw the block diagram and describe the instrumentation and applications of IR spectrophotometer. (12)

   (b) What are the factors affecting the position and intensity of UV-visible absorption bands? (8)

2. (a) What are the various factors affecting the chemical shift? (5)

   (b) How will you identify the organic and inorganic compounds using ESR spectroscopy? Explain it with suitable examples. (10)

   (c) Write note on NMR shift reagents. (5)

3. (a) Draw and explain the TGA curve of calcium oxalate monohydrate. (7)

   (b) Explain the principles and applications of calorimetry. (6)

   (c) Explain the factors which affect TGA and DTA curves. (7)

4. (a) Discuss the colorimetric determination of chromium. (10)

(b) What quantity is measured in

(i) thermo gravimetric analysis

(ii) differential thermal analysis

(iii) thermometric application. (5)

(c) Explain how DTA differs from TGA. (5)

5. (a) How will you determine phosphate by nephalometrically? (8)

(b) Explain the principles and applications of ORD and CD. (12)

6. (a) Describe an experiment for the separation of a mixture of chloride, bromide, iodide ions by paper chromatography. (9)

(b) Write notes on :

(i) Meta stable ion (3)

(ii) Mclafferty rearrangement (4)

(iii) Octant rule. (4)

7. (a) Explain whether the following errors are determinate or indeterminate, if the error is indeterminate tell whether it is methodic/operative or instrumental and how it could be eliminated?

(i) Analytical weights are corroded

(ii) The analyst unknowingly spatters some solution from a flask during the titration

(iii) A sample picks up moisture during a weighing

(iv) A reagent is used which contains some of the substance being determined

(v) The burette is misread once. (10)

(b) Discuss the principles and applications of chrono techniques. (10)

8. (a) The following values were obtained for the determination of calcium in a sample of hard water : 7.6, 7.3, 7.1, 7.0 and 6.2 mg/lit. Should the last value 6.2 he rejected or retained? (Q. critical = 0.717) (6)

(b) Quinine was determined by measuring the fluorescence intensity, in 1m H_2SO_4 solution. Standard solution of quinine gave the following fluorescence value.

[Quinine] :	0.0	0.10	0.20	0.30	0.40
Fluorescence :	0.0	5.20	9.90	15.30	19.10

Calculate the correlation coefficient (r) . (4)

(c) What is student-t test? Explain its importance. (4)

(d) Discuss the principles and applications of polarography. (6)

DE–4147 **21**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

ORGANIC CHEMISTRY – II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

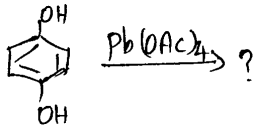
Each question carry 20 marks.

$(5 \times 20 = 100)$

1. (a) Discuss the mechanisms of Wolff rearrangement and Stern and Favorski rearrangement. (12)

   (b) Write oppenaur oxidation with examples. (4)

   (c) Write the products for following reactions. (4)

   (i) 

   (ii)

2. (a) Discuss the types of isomerism exhibited by spiranes. (6)

   (b) Explain asymmetric synthesis with example. (6)

   (c) Discuss Chan-Ingold-Prelog rules for assigning the configuration of a chiral carbon. (8)

3. (a) Discuss the mechanism of E_2 and cis-elimination reactions. (10)

(b) Explain Neighbouring group participation with example. (6)

(c) Discuss the mechanism of Pschorr rearrangement.(4)

4. (a) Discuss Paterno-Buchi reaction and photo reduction reaction with example. (10)

(b) Explain the following reactions with example.

(i) Chemotropic reaction

(ii) Sigmatropic rearrangement. (10)

5. (a) Establish the structure of cholesterol. (10)

(b) Discuss the biosynthetic route for cholesterol. (5)

(c) Write the synthesis of Vitamin B_1. (5)

6. (a) What are activating groups? Illustrate with examples. (5)

(b) Explain Umpolung synthesis with example. (7)

(c) Discuss in detail the optical isomerism of compounds containing one asymmetric carbon. (8)

7. (a) Explain the following reactions with examples.

(i) Gomberg

(ii) Sandmayer

(iii) Hunsdicker. (9)

(b) Discuss the detection and stability of free radicals. (8)

(c) What is Barton reaction? Explain how this reaction is useful in functionalising angular methyl group of steriods. (3)

8. (a) Establish the structure of progesterone and Androsterone. (8)

(b) How will you synthesis vitamin B_2? (8)

(c) Write one example for the oxidation reaction using chromic acid, DMSO and MnO_2 as oxidising agents. (4)

DE–4148 **22**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

INORGANIC CHEMISTRY – II

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Write down the IUPAC nomenclature of $[Ni(Co)_4]$ and $[Ag(NH_3)_2]Cl$. (4)

   (b) Explain the structure of complexes with co-ordination number 5 with suitable examples. (6)

   (c) Discuss the electron delocalisastion in metal complexes. Describe the features of MOT. (10)

2. (a) Explain chelate effect. (5)

   (b) Discuss on the site preferences in spinnels. (5)

   (c) Explain the inner sphere and outer sphere processes. (10)

3. (a) Explain the significance of Racah parameters. Explain how 10Dq can be determined for Ni(II) octahedral complexes. (8)

   (b) Explain the basic principles of EPR. Discuss the splitting in bis (Salicylaldimine copper (II)). (8)

   (c) Outline the group vibrations in the structural elucidation of metal complexes of urea. (4)

4. (a) Explain the reactivity, magnetic property and stability of ferrocene. (7)

(b) Discuss Dewar-Chatt approach to bonding in olefins. (7)

(c) Explain the mechanism of Ziegler-Natta Polymerisation. (6)

5. (a) Discuss on Jahn-Teller distortion theorem. (6)

(b) Explain the synthesis of macrocyclic ligands. (6)

(c) Describe the hole-formalism. (8)

6. (a) Distinguish between trans effect and trans influence. (7)

(b) Discuss the structure and biological activity of vitamin B_{12}. (8)

(c) Write a note on chelating agents in medicine. (5)

7. (a) Write about different types of copper proteins. (6)

(b) Derive a relation between the stepwise and overall stability constants of complexes. How will you determine the stability constant of coordination compounds? (8)

(c) Write about the fixation of nitrogen in viro systems. (6)

8. (a) The compound $Ni(en)_2(CN)_2$ can be formulated either as $[Ni(en)_2(CN)_2]$ or $[Ni(CN)_2(en)_2]$. Explain by CFT interpretation for the structural assignments. (5)

(b) Write notes on :

(i) aquation in octahedral complexes.

(ii) antiferromagnetism.

(iii) fluxinal molecules. (9)

(c) Discuss on the geometrical and optical isomerisms in square planar complexes. (6)

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

PHYSICAL CHEMISTRY – II

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Explain statistical interpretation of third law. (8)

   (b) Derive Onsager reciprocal relations. (12)

2. (a) A hydrogen atom is exposed to an electric field of strength F applied in the Z – direction. Calculate the first order for the ground state of atom (stark effect) (5)

   (b) Discuss in detail about Pauli's exclusion principle. (7)

   (c) Verify that $\psi = A \sin kx + B \cos kx$ is a general solution for a particle in a one-dimensional box having infinite potential walls. (8)

3. (a) Explain why a set of numbers cannot form a group by the process of division. (8)

   (b) Construct the multiplication table for the symmetry – operations of water molecule. (6)

   (c) Show that the character system for $C_2\gamma$ obey the rules of irreducible representation. (6)

4. (a) Write briefly on primary salt effect and secondary salt effect. (12)

(b) Discuss in detail about Michaelis – Menton equation. (8)

5. (a) Discuss Langmuir –Hinshelwood mechanism of Uni and bi molecular surface reactions. (10)

(b) Explain Langmuir – Riedal mechanism with suitable examples. (10)

6. (a) Give an account of translational, rotational and vibrational partition functions. (8)

(b) State and explain Fermi – Dirac distribution law. (8)

(c) Define Bose – Einstein distribution law. (4)

7. (a) Discuss the great Orthogonality theorem. (8)

(b) How does group theory apply to IR and Raman spectra? (12)

8. (a) Give an account of adsorption of gases on solids. (8)

(b) Write notes on Freudlich and Langmuir isotherms. (12)

DE–4150 **24**

DISTANCE EDUCATION

M.Sc. (Chemistry) DEGREE EXAMINATION, MAY 2012.

APPLIED CHEMISTRY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carry 20 marks.

$(5 \times 20 = 100)$

1. (a) Write a note on the causes and consequences of thermal pollution. (12)

   (b) Write briefly on biological and advanced methods of wasteswater treatment. (8)

2. (a) Explain the important aspects of electroplating. (8)

   (b) Give the general account of classification of corrosion and corrosion control methods. (12)

3. (a) Discuss the classification of polymers in detail. (12)

   (b) Write notes on

      (i) Fire retardant polymers

      (ii) Thermally stable polymers. (8)

4. (a) Write briefly on the synthesis of nonomaterials by chemical methods. (12)

   (b) Write notes on DSC and TG. (8)

5. (a) Give the importance of the following terms used in internet with detail.

(i) Bandwidth

(ii) Dial up service

(iii) ISDN

(iv) HTTP

(v) WWW (10)

(b) Give an account of internet facility in relevance to computer applications in experimental aspects of chemistry. (10)

6. (a) Explain the formation of acid rain and its adverse effects. (8)

(b) Discuss the principle and applications of electrolessplating and electroforming. (12)

7. (a) Discuss the following methods of polymerization

(i) Solution polymerization

(ii) Bulk polymerization (10)

(b) Discuss the important aspects of the following microscopic characterisation techniques of nanomaterials like AFM and TEM. (10)

8. (a) Give an account of various internet service providers. (10)

(b) Describer the causes for air pollution and also the methods to minimize it. (10)